Characterization of Millimeter-Wave RF Components Using a Low-Cost SDR Platform for 5G NR Systems

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***Abstract*—The characterization of millimeter-wave (mmWave) radio frequency (RF) hardware is a prerequisite for the reliable deployment of 5G systems, yet the metrology-grade instrumentation conventionally required for this task remains prohibitively costly and, in many cases, altogether unavailable. This paper presents the experimental characterization of mmWave passive and active RF components, carried out using a low-cost, reconfigurable Software-Defined Radio (SDR) based intermediate frequency (IF) loopback platform. Passive devices are evaluated through a magnitude-based root-mean-square (RMS) power detection scheme calibrated at IF, enabling the extraction of frequency-dependent insertion loss beyond the reach of conventional instrumentation. Active devices, in turn, are characterized dynamically through a self-referenced Error Vector Magnitude (EVM) methodology that intercepts uncompensated baseband In-Phase and Quadrature samples to expose the Amplitude-to-Amplitude and Amplitude-to-Phase distortion trajectories of a mmWave power amplifier under realistic 5G NR CP-OFDM excitation. The proposed methodology is experimentally validated across the mmWave band, demonstrating that comprehensive passive and active hardware characterization can be achieved without reliance on dedicated mmWave test equipment.**

***Index Terms*—5G New Radio, millimeter-wave, Software-Defined Radio, hardware characterization, IF-loopback, error vector magnitude, hardware dispersion.**

I. Introduction

Fifth-generation (5G) New Radio (NR) systems increasingly rely on millimeter-wave (mmWave) frequency bands to deliver the massive bandwidth and high-order modulation schemes, such as 256-QAM, required for multi-gigabit data rates [1], imposing stringent validation requirements on every element of the radio frequency (RF) chain, passive and active alike. For passive components, ensuring compliance is fundamentally a matter of experimental validation: once a device such as a filter or a power splitter has been fabricated, its frequency response, insertion loss, and bandwidth must be verified against the original design before it can be trusted within a larger system [2]. Manufacturing tolerances and packaging effects at mmWave frequencies render this verification step particularly critical, as even minor deviations can shift the measured response away from its intended target [3]. Active components, in turn, are subject to a further layer of scrutiny: the high peak-to-average power ratio (PAPR) inherent to the Cyclic Prefix Orthogonal Frequency-Division Multiplexing (CP-OFDM) waveforms employed in 5G NR readily drives power amplifiers (PAs) into compression, giving rise to nonlinear Amplitude-to-Amplitude (AM-AM) and Amplitude-to-Phase (AM-PM) distortion that degrades the Error Vector Magnitude (EVM) and induces out-of-band spectral regrowth [4], [5]. Consequently, characterizing these AM-AM/AM-PM trajectories together with the static response of passive devices constitutes an indispensable step toward reliable mmWave hardware, as they yield the PA's behavioral memory model underlying digital pre-distortion (DPD) techniques used to mitigate such distortion [6].

Such characterization conventionally relies on metrology-grade instrumentation, including Vector Network Analyzer (VNA) and Vector Signal Analyzer (VSA) devices, whose cost increases drastically with operating frequency [7]. This barrier becomes even more restrictive at mmWave frequencies, where suitable test equipment is frequently unavailable altogether, leaving research groups unable to verify their designs through conventional means. Under such conditions, neither the frequency response of a fabricated passive component nor the non-linear behavior of an active amplifier under realistic modulation can be directly measured, leaving the corresponding designs effectively unverified against their

intended specifications.

This work addresses this gap by presenting a complete, low-cost methodology for the experimental characterization of mmWave passive and active RF components, without relying on costly or otherwise unavailable commercial instrumentation. The contributions of this paper are the following. (i) passive components are evaluated through a magnitude-based root-mean-square (RMS) power detection scheme calibrated at intermediate frequency (IF), enabling accurate extraction of insertion loss and frequency response well beyond conventional instrumentation limits. (ii) Active components are characterized dynamically through a self-referenced EVM-based methodology that extracts uncompensated baseband I/Q samples to map the AM-AM and AM-PM distortion trajectories of mmWave power amplifiers under realistic 5G NR CP-OFDM excitation. (iii) Both characterization approaches are implemented and experimentally validated on an accessible, heterodyne Software-Defined Radio (SDR) IF-loopback testbed, demonstrating that comprehensive mmWave hardware verification can be achieved without dedicated mmWave instrumentation.

II. Testbed Architecture

The proposed characterization platform is engineered as a reconfigurable IF architecture shown in Fig. 1 and physically implemented in Fig. 2. In its closed-loop form, this architecture supports dynamic 5G NR stress testing of active devices, capturing its high-power non-linear behavior. For the characterization of passive components, the same architecture is instead operated open-loop. The down-conversion path is left inactive and power is measured directly at the Device Under Test (DUT) output, in the configuration detailed in Section III-A (Fig. 4).

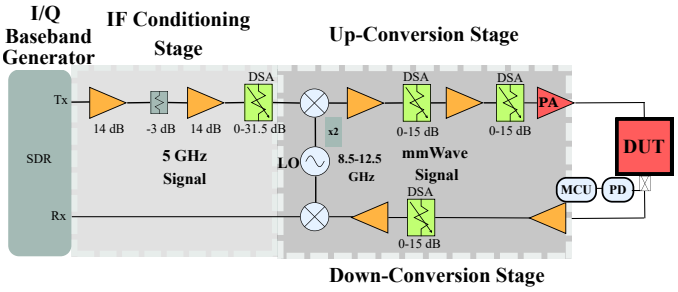


Fig. 1: Block diagram of the reconfigurable SDR-based characterization testbed.

As illustrated in the block diagram, the signal path begins at a baseband in-phase/quadrature (I/Q) generation stage, whose IF output is routed through a conditioning stage before entering the up-conversion block. Within this block, an internal amplification and step-attenuation chain, driven by a variable-frequency local oscillator, translates the signal to the mmWave band and delivers it to the DUT, which is left unspecified in the diagram to emphasize that the architecture accommodates arbitrary passive or active mmWave components. At the DUT output, a directional coupler splits the signal between a dedicated power-detection branch and a down-conversion path that re-enters the same mixing block, where an internal low-noise amplification and step-attenuation chain returns the signal to its original IF before it is delivered back to the baseband I/Q stage for acquisition.

The corresponding physical implementation, labeled A–H in Fig. 2, shows this architecture as built for the present characterization process. The baseband I/Q generation stage (A) is realized using the ADALM-PLUTO Software-Defined Radio (SDR) module [8]. Its IF output is conditioned by a dedicated amplification board (B), built around two BGU7258 low-noise amplifiers and a PE43711 digital step attenuator [9], [10], before entering the mixer (D) [11]. Within the mixer, the signal is combined with the output of a variable-frequency local oscillator (C) [12], internally multiplied by two, to produce a mmWave signal centered at 26 GHz. This signal is delivered to the DUT, in this case a mmWave bandpass filter (E) [13], whose output is routed through a directional coupler (F) [14]. One coupler branch feeds an RMS power detector (G) [15], mounted on its dedicated evaluation board, whose voltage output is read by a microcontroller (H) [16] through an ADC pin to derive and register the corresponding power value, while the other branch returns to the mixer (D) for down-conversion back to the original IF. The resulting signal is finally routed back to the SDR (A), which captures and digitizes it for subsequent processing.

In closed-loop operation, the SDR (A) transmits a 5G NR CP-OFDM waveform (256-QAM, 20 MHz bandwidth) at the nominal IF, which stresses the active DUT before returning through the mixer's receive path to the SDR. Beyond this campaign, Fig. 3 summarizes what each configuration yields: insertion loss and frequency response for passive components (open-loop), and AM-AM and AM-PM trajectories with the resulting distorted output for active components (closed-loop).

III. Experimental Procedure

To address the metrological requirements of mmWave RF front-ends, the experimental methodology is structured into two phases. First, a static characterization is performed to accurately isolate mmWave insertion losses. Second, a dynamic closed-loop evaluation establishes the system's linearity boundaries and extracts real-time non-linear distortion profiles.

A. Characterization of Static Components

In order to extract insertion loss profiles, an RMS power detector must be validated to guarantee that its analog output tracks the true RF power. To this end, the internal transmission gain of the SDR was swept from -50 dB to 0 dB while radiating an unmodulated 5 GHz CW tone, and the detector's DC voltage output was simultaneously sampled by the onboard microcontroller's ADC and cross-referenced against the absolute power measured by a calibrated, 12.5 GHz-range Spectrum Analyzer (SA). This procedure establishes the sensor's transfer function, converting raw ADC voltage readings into calibrated RF power values in dBm. Because

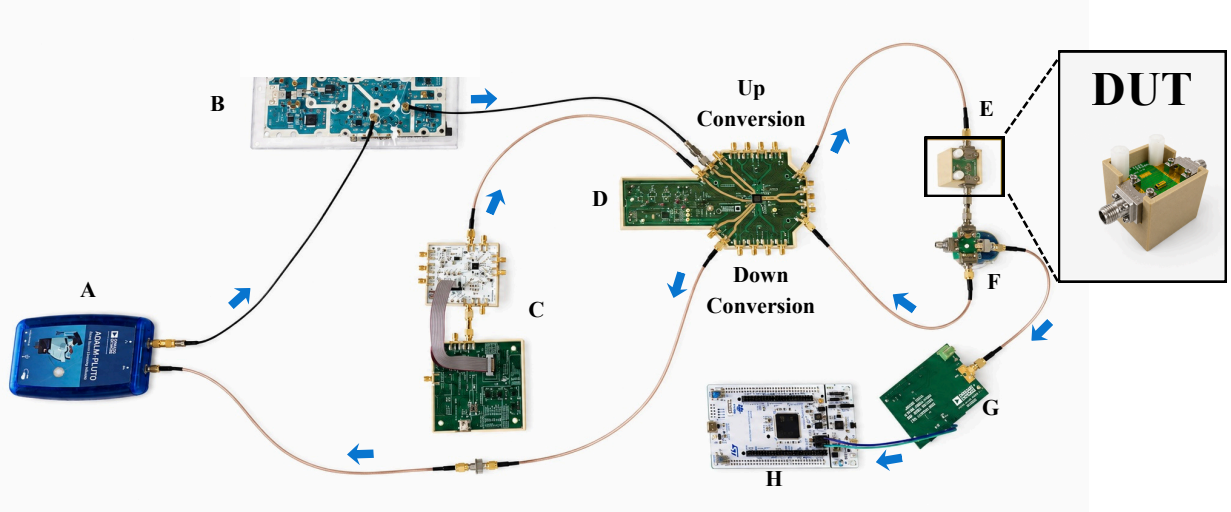


Fig. 2: Physical implementation of the characterization testbed. The labels A–H denote the specific components utilized for the 26 GHz evaluation campaign.

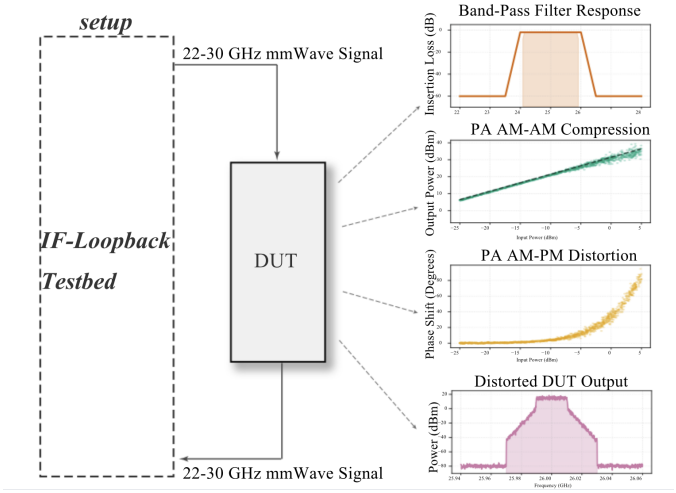


Fig. 3: Reconfigurable testbed operation: static RMS-based and dynamic I/Q-loopback modes.

this reference SA is limited to 12.5 GHz, the resulting curve is extrapolated to the 26 GHz DUT interface; since insertion loss is a difference between two readings from the same detector, any intercept offset cancels exactly, leaving only the residual slope mismatch between 5 GHz and 26 GHz, bounded to $\sim$1% by the detector's specified flat response [15]. The same datasheet specifies a usable input range from -37.8 to 5 dBm at 26 GHz within ± 1 dB linearity.

Because the insertion loss of a passive DUT is defined relative to the power delivered by the up-conversion stage, this reference power must itself be established before the DUT is inserted into the signal path. To this end, the mixer's conversion gain is first obtained by connecting the power detector directly to its output, in the same physical configuration later used to accommodate the DUT. At this and all subsequent mmWave interfaces, a barrel adapter is used in place of coaxial cabling to interconnect the mixer, the DUT, and the power detector, minimizing external losses that would otherwise be conflated with those of the DUT itself. Both the reference measurement and the subsequent DUT characterization are repeated across a 22–30 GHz sweep obtained by varying the local oscillator frequency, allowing insertion loss to be resolved as a function of frequency rather than at a single operating point.

For each frequency $f \in [22, 30]$ GHz, the insertion loss $\mu_{IL}(f)$ is computed as the average power difference between the DUT and reference measurements:

$$\mu_{IL}(f) = \frac{1}{N} \sum_{n=1}^{N} \left(P_{\mathrm{DUT},n}(f) - P_{\mathrm{mix},n}(f) \right) \qquad (1)$$

where $P_{\mathrm{DUT},n}(f)$ and $P_{\mathrm{mix},n}(f)$ are, respectively, the n-th power sample measured at the mixer output and at the DUT output using the RMS detector's DC voltage reading, and $N = 50$ is the number of independent samples acquired at each frequency point, chosen as a conservative sample size sufficient to yield a stable estimate of $\mu_{IL}(f)$.

The physical test configuration used for this characterization, in which the DUT is inserted between the up-conversion mixer and the RMS power detector, is shown in Fig. 4; the mixer reference measurement is obtained under the same configuration with the DUT block removed, connecting the detector directly to the PA output.

B. Dynamic System Baseline

Once the static characterization is completed, the next step is to capture the non-linear trajectories of active components, for which the testbed's baseline must first be established. Using

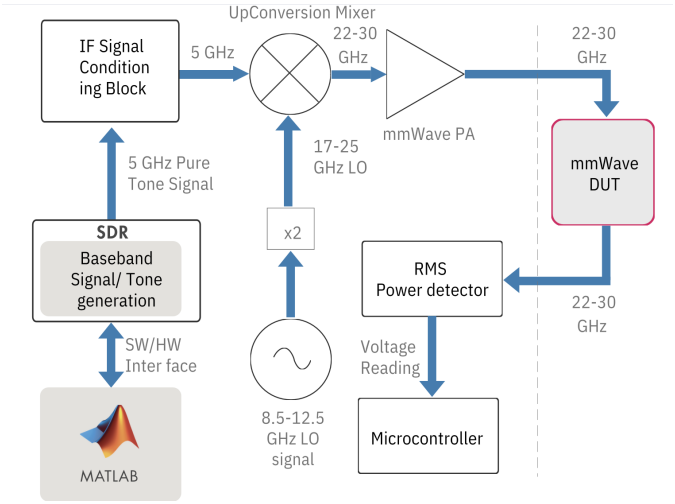


Fig. 4: Static characterization test setup.

the dynamic 5G loopback mode, a two-dimensional sweep of the SDR's transmission and reception gains is executed, spanning -45 to -5 dB and 0 to 30 dB, respectively, in 1 dB steps, to map the system's operational EVM matrix. This sweep identifies the gain configuration yielding the lowest achievable EVM, establishing an optimal linear operating region and ensuring that any subsequent signal degradation is primarily attributed to the mmWave DUT rather than to the baseband transceiver itself.

Once this baseline is secured, with the DSA preceding the PA fixed at a nominal 15 dB attenuation, the active mmWave power amplifier is systematically stressed by incrementally reducing this attenuation, thereby injecting progressively higher power into the PA.

C. Transient I/Q Extraction for AM-AM and AM-PM Profiling

To physically map the AM-AM and AM-PM compression trajectories mentioned above, uncompensated baseband samples must be intercepted. In modern radio architectures, standard demodulation algorithms apply channel equalization and common phase error (CPE) compensation that mathematically mask physical hardware transients and thermal memory effects.

To bypass these digital corrections, the internal 5G Toolbox demodulation function was modified. As conceptualized in Fig. 5, the continuous waveform is retrieved at an intermediate node, immediately following frame synchronization but strictly prior to equalization. Extracting these raw In-Phase and Quadrature (I/Q) samples enables the direct visualization of the active device's AM-AM and AM-PM distortion trajectories; the scatter observed around these trajectories as the amplifier approaches saturation is characteristic of the onset of memory effects. 256-QAM constellations are included as a visual reference of signal quality degradation.

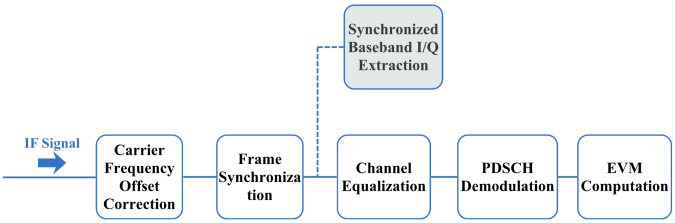


Fig. 5: Receiver-side Digital Signal Processing pipeline for 5G NR EVM measurement based on 3GPP TS 38.104. The highlighted extraction node indicates where uncompensated I/Q samples are retrieved to isolate transient AM-AM distortions.

IV. Experimental Results

The experimental validation of the SDR-based IF-loopback platform was carried out across the 22–30 GHz mmWave band. The results are organized into the static characterization of the passive DUT and the dynamic evaluation of the active PA under 5G NR stress, including the extraction of its AM-AM and AM-PM distortion trajectories.

A. Static Characterization of the Passive DUT

To ensure the testbed's measurement transparency, the RMS power detector described in Section III-A was first calibrated against a reference SA, as shown in Fig. 6. The sensor exhibits a highly linear response, with a slope of 28.82 mV/dB and a coefficient of determination of $R^2 = 0.9997$, confirming that its DC voltage output accurately represents the RF power without introducing sensor-induced non-linearities.

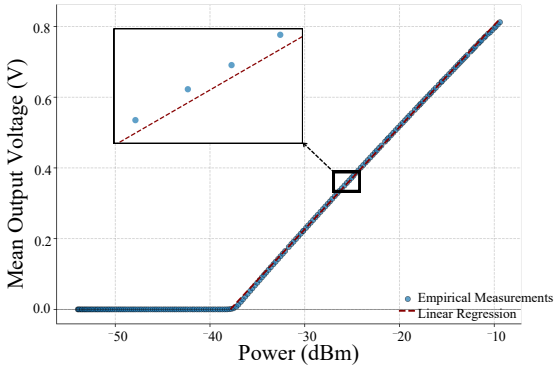


Fig. 6: RMS power detector transfer function.

Subsequently, the insertion loss $\mu_{IL}(f)$ of the passive mmWave DUT was quantified across $f \in [22, 30]$ GHz following the procedure outlined in Section III-A.

Fig. 7 presents the output power measured at the mixer, prior to DUT insertion, overlaid with the power subsequently measured at the output of the mmWave bandpass filter, across the same 22–30 GHz sweep, with shaded regions indicating the sample-to-sample variation at each frequency point. The mixer's own frequency response is not perfectly flat: a pronounced attenuation dip reaching approximately -20 dBm is observed near 28.3 GHz that is absent from the mixer IC's

own datasheet response [11], suggesting a board-level origin; this stays well above the detector's -37.8 dBm sensing floor, so measurement validity in this sub-band is unaffected. This illustrates the purpose of the proposed testbed, uncovering the true response of mmWave hardware as deployed.

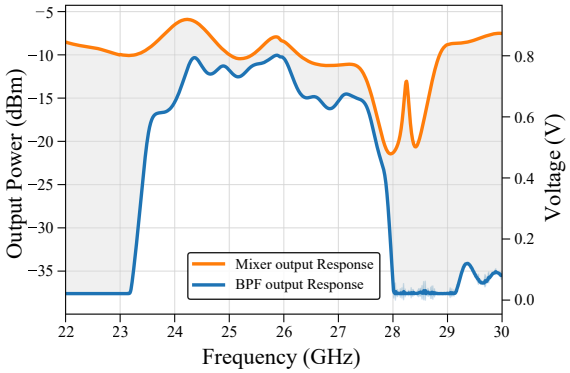


Fig. 7: Mixer and filter output power across the mmWave sweep.

Subtracting the filter output trace from the mixer reference trace at each frequency point yields the insertion loss profile shown in Fig. 8. By capturing multiple samples, the attenuation of the device was isolated from thermal noise and LO phase variations, with the narrow shaded band confirming the low sample-to-sample variability of the measurement. The filter exhibits a sharply defined passband between 24.25 and 28 GHz. These results show that the platform can measure mmWave insertion loss without a commercial VNA.

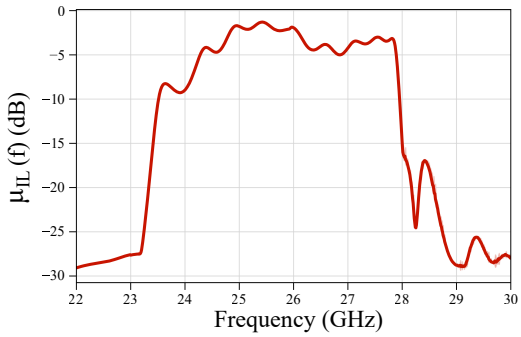


Fig. 8: Insertion loss profile of the mmWave bandpass filter.

B. Dynamic 5G NR Linearity Boundaries

With the static baseline verified, the platform was reconfigured into the dynamic IF-loopback mode. Fig. 9 presents the EVM heatmap from the two-dimensional gain sweep of Section III-B (green: low EVM, red: high EVM). The black marker identifies the lowest EVM, 1.27%, obtained at a transmission gain of -25 dB and a reception gain of 18 dB. This minimum-EVM point serves as the baseline for the PA tests.

With this baseline fixed, the DSA preceding the PA was progressively de-attenuated to inject increasing power into the

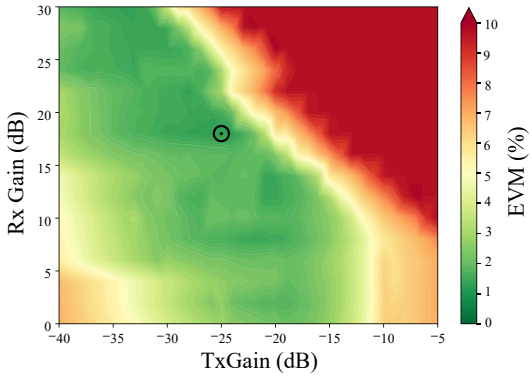


Fig. 9: EVM heatmap across the Tx/Rx gain sweep. The black marker denotes the operating point.

amplifier. Fig. 10 shows the resulting EVM as a function of the DSA setting value, revealing a clear degradation in signal quality as the PA is driven from its linear regime toward saturation.

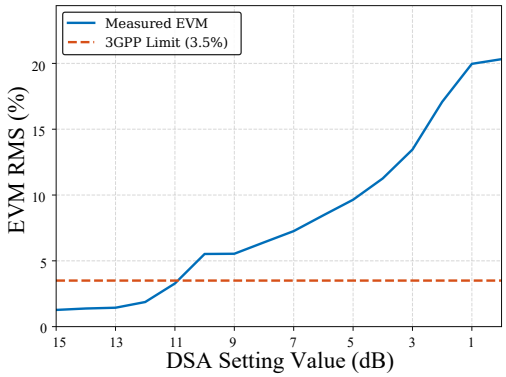
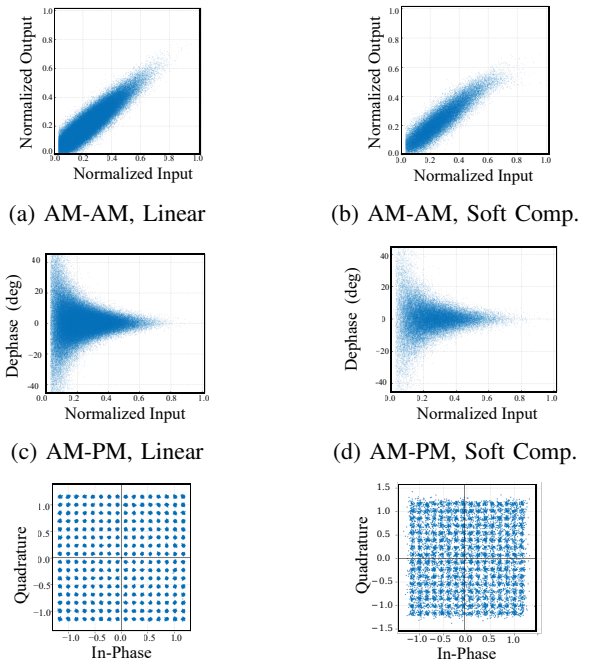


Fig. 10: EVM degradation as power is injected into the PA.

C. Transient Memory Effects: AM-AM and AM-PM Trajectories

To explain the EVM degradation observed in Fig. 10, uncompensated I/Q samples were extracted at two representative points: a linear operating condition (EVM = 1.27%, Section IV-B) and an intermediate soft-compression point (EVM = 5.52%). The resulting AM-AM and AM-PM trajectories, together with their 256-QAM constellations, are shown in Fig. 11. The AM-AM trace clearly captures the expected nonlinear behavior: it closely follows the ideal linear reference under the low-power condition and exhibits a discernible gain reduction at high input amplitude under soft compression, consistent with the onset of PA non-linearity; the constellation correspondingly degrades from a clean 256-QAM grid to a visibly dispersed pattern. The AM-PM trace, in contrast, exhibits a pronounced phase deviation at low input amplitudes in both operating conditions, with the baseband generator's own phase noise dominating the phase estimate in this region. At high amplitude, where compression actually occurs and this noise contribution is negligible, the high PAPR of the OFDM

waveform instead yields comparatively few samples, limiting the statistical visibility of the underlying trend; this effect still allows compression to be discerned in the AM-AM trace, but prevents a clear identification of the corresponding AM-PM trend. Future work will examine whether stages other than the DUT contribute to this behavior, in particular the SDR, which acts as both transmitter and receiver and may introduce phase noise, receiver saturation, or coupling between its transmit and receive paths. Nonetheless, the clear AM-AM divergence between the two operating points confirms the platform's ability to detect PA compression from uncompensated baseband I/Q samples.



(a) AM-AM, Linear (b) AM-AM, Soft Comp.

(c) AM-PM, Linear (d) AM-PM, Soft Comp.

(e) Constellation, Linear (f) Constellation, Soft Comp.

Fig. 11: Measured AM-AM, AM-PM, and 256-QAM constellation at the linear and soft-compression operating points.

V. Conclusion

This paper presented a low-cost, SDR-based IF-loopback methodology for the experimental characterization of mmWave passive and active RF components, addressing the practical inaccessibility of metrology-grade instrumentation at these frequencies. Using a magnitude-based RMS power detection scheme calibrated at IF, the platform isolated the insertion loss profile of a mmWave bandpass filter while incidentally revealing a board-level attenuation feature near 28.3 GHz that the mixer datasheet does not show Using a self-referenced EVM methodology, the platform further identified a minimum-EVM baseband operating point (EVM = 1.27%) and, by extracting uncompensated I/Q samples, clearly captured the onset of AM-AM compression in a PA driven toward saturation, while also revealing amplitude-dependent limitations in the AM-PM extraction. Together, these results validate the proposed testbed as an accessible alternative for mmWave hardware verification; future work will address the amplitude-dependent limitations of the AM-PM extraction, and pursue empirical modeling of amplifier memory effects for DPD. The insertion-loss characterization will extend to additional mmWave components as equipment access permits, to further validate the platform.

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